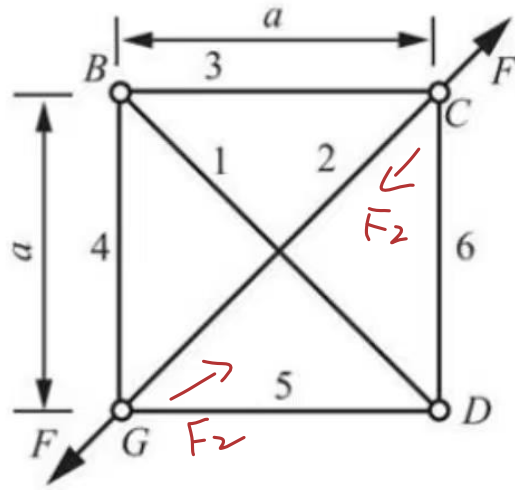


材料为线弹性，拉压刚度为 EA 的超静定桁架及其承载情况如图所示，试用卡氏第二定理求各杆的轴力。



为一次超静定，解除杆2约束。

$$\text{对 G 点 } F_4 = F_5 = F_3 = F_6 = \frac{\sqrt{2}}{2} (F - F_2)$$

$$F_1 = F_2 - F$$

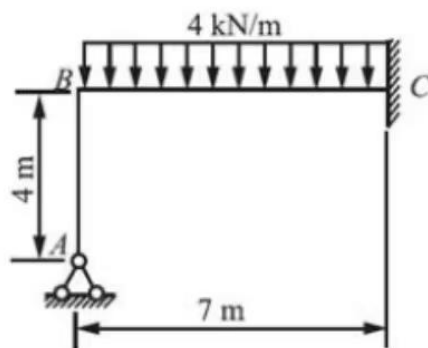
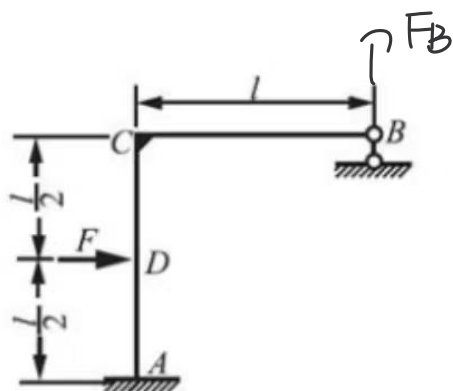
$$V_\varepsilon = \sum \frac{F_i l_i}{2EA} = \frac{1}{2EA} \left(\sqrt{2}a \cdot (F - F_2)^2 + \sqrt{2}a F_2^2 + 4a \left[\frac{\sqrt{2}}{2} (F - F_2) \right]^2 \right)$$

$$A = \frac{\partial V_\varepsilon}{\partial F_2} = \frac{a}{2EA} [-2(2 + \sqrt{2})F + 2(2 + 2\sqrt{2})F_2] = 0$$

$$\therefore F_2 = \frac{\sqrt{2}}{2} F$$

$$\therefore \begin{cases} F_1 = \frac{\sqrt{2}-2}{2} F \\ F_2 = \frac{\sqrt{2}}{2} F \end{cases} \quad F_3 = F_4 = F_5 = F_6 = \frac{\sqrt{2}-1}{2} F$$

材料为线弹性，弯曲刚度为EI的各超静定刚架分别如图所示，不计轴力和剪力的影响，试用卡氏第二定理求各刚架的支反力。



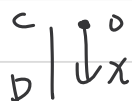
对于左图

对BC



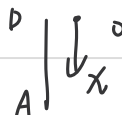
$$M_{BC}(x) = x \cdot F_B$$

对CD



$$M_{CD}(x) = l \cdot F_B$$

对DA



$$M_{DA}(x) = l \cdot F_B - F \cdot x$$

$$A_B = 0 \quad \therefore \frac{1}{EI} \left[\int_0^l M_{BC}(x) \frac{\partial M_{BC}(x)}{\partial F_B} dx + \int_0^{\frac{l}{2}} M_{CD}(x) \frac{\partial M_{CD}(x)}{\partial F_B} dx + \int_0^{\frac{l}{2}} M_{DA}(x) \frac{\partial M_{DA}(x)}{\partial F_B} dx \right]$$

$$= \frac{1}{3} F_B l^3 + \frac{1}{2} F_B l^3 + \frac{1}{2} F_B l^3 - \frac{1}{8} F_B l^3 = 0$$

$$\therefore F_B = \frac{3}{32} F$$

对A点

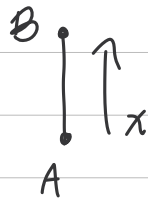
$$F_{Ax} + F = 0 \quad \therefore F_{Ax} = -F$$

$$F_{Ay} + F_B = 0 \quad F_{Ay} = -\frac{3}{32} F$$

$$M_A = \frac{Fl}{2} - \frac{3Fl}{32} = \frac{13}{32} Fl$$

对右图.

对AB段



$$M_{AB}(x) = -F_{Ax} \cdot x.$$

对BC段.



$$M_{BC}(x) = F_{Ay} \cdot x - 4F_{Ax} - 2x^2$$

$$A_{Ax} = A_{Ay} = 0$$

$$\frac{1}{EI} \int \frac{\partial M}{\partial F_{Ay}} ds = 0 \quad \Rightarrow \int_0^4 M_{AB}(x) dx + \int_0^7 (F_{Ay}x - 4F_{Ax} - 2x^2) \cdot x dx = 0$$

$$\therefore F_{Ay} \frac{343}{3} - F_{Ax} \cdot 98 - \frac{2401}{2} = 0. \quad (1)$$

$$\frac{1}{EI} \int \frac{\partial M}{\partial F_{Ax}} ds = 0 \quad \therefore \int_0^4 -F_{Ax} x (-x) dx + \int_0^7 (F_{Ay}x - 4F_{Ax} - 2x^2) (-x) dx = 0$$

$$\therefore \frac{400}{3} F_{Ax} - 98 F_{Ay} + \frac{2744}{3} = 0. \quad (2)$$

$$\begin{cases} (1) \\ (2) \end{cases} \Rightarrow F_{Ax} = 2.32 \text{ kN (向右)}$$

$$F_{Ay} = 12.49 \text{ kN (向上)}$$

$$F_{Ax} + F_{Cx} = 0 \quad \therefore F_{Cx} = -2.32 \text{ kN.}$$

$$F_{Ay} + F_{Cy} - (4 \times 7) = 0 \quad \therefore F_{Cy} = 15.51 \text{ kN.}$$

$$M_c + F_{Ay} \cdot 7 + F_{Ax} \cdot 4 - 4 \times 7 \times 3.5 = 0$$

$$\therefore M_c = 1.32 \text{ kN} \cdot \text{m.}$$

综上

$$\begin{cases} F_{Ax} = 2.32 \text{ kN} & F_{Cx} = -2.32 \text{ kN} \\ F_{Ay} = 12.49 \text{ kN} & F_{Cy} = 15.51 \text{ kN} \\ & M_c = 1.32 \text{ kN} \cdot \text{m} \end{cases}$$